Extra Dessert #3: HF Digital Modes

FT8, FT4, WSPR, JT65/Q65, PSK31, PACTOR, RTTY, ALE -- what each does and how they differ.

Group E2E · 13 pool questions · Self-study reference · Practice exams at eham.net/exams

Modulation Basics

Question	Answer	Q
Modulation used for data emissions below 30 MHz	**FSK** (frequency shift keying)	E2E01
Direct FSK vs audio FSK	**Direct FSK modulates the transmitter VFO**; audio FSK feeds audio tones into the mic input and uses SSB	E2E11

WSJT-X Modes (FT8, FT4, JT65, Q65, FST4, WSPR)

Topic	Answer	Q
WSJT-X timing synchronization	**Synchronization of computer clocks** (typically via NTP / Internet time)	E2E02
FT8 transmission cycle length	**15 seconds**	E2E06
What the “4” in FT4 means	**Four-tone continuous-phase frequency shift keying**	E2E03
FST4 characteristics	Four-tone Gaussian FSK + variable transmit/receive periods + seven different tone spacings – **all of the above**	E2E04
Q65 vs JT65	**Multiple receive cycles are averaged** – that is how Q65 digs weaker signals out of noise	E2E07
Mode that does NOT support keyboard-to-keyboard	**WSPR** (Weak Signal Propagation Reporter – beacon mode only, no QSO)	E2E05
Narrowest bandwidth mode	**FT8**	E2E10

PSK31, PACTOR, RTTY, ALE

Question	Answer	Q
Variable-length character coding	**PSK31** (uses varicode – common letters get short codes)	E2E09
Binary file transfer over HF digital	**PACTOR**	E2E08
Highest data throughput in clear conditions	**PACTOR IV**	E2E13
How ALE stations establish contact	**ALE constantly scans a list of frequencies, activating the radio when the designated call sign is received**	E2E12

Dessert 3 Question Bank (E2E)

E2E01. Which of the following types of modulation is used for data emissions below 30 MHz?
A. DTMF tones modulating an FM signal.
B. FSK.
C. Pulse modulation.
D. Spread spectrum.

E2E02. Which of the following synchronizes WSJT-X digital mode transmit/receive timing?
A. Alignment of frequency shifts.
B. Synchronization of computer clocks.
C. Sync-field transmission.
D. Sync-pulse timing.

E2E03. To what does the “4” in FT4 refer?
A. Multiples of 4 bits of user information.
B. Four-tone continuous-phase frequency shift keying.
C. Four transmit/receive cycles per minute.
D. All these choices are correct.

E2E04. Which of the following is characteristic of the FST4 mode?
A. Four-tone Gaussian frequency shift keying.
B. Variable transmit/receive periods.
C. Seven different tone spacings.
D. All these choices are correct.

E2E05. Which of these digital modes does not support keyboard-to-keyboard operation?
A. WSPR.
B. RTTY.
C. PSK31.
D. MFSK16.

E2E06. What is the length of an FT8 transmission cycle?
A. It varies with the amount of data.
B. 8 seconds.
C. 15 seconds.
D. 30 seconds.

E2E07. How does Q65 differ from JT65?
A. Keyboard-to keyboard operation is supported.
B. Quadrature modulation is used.
C. Multiple receive cycles are averaged.
D. All these choices are correct.

E2E08. Which of the following HF digital modes can be used to transfer binary files?
A. PSK31.
B. PACTOR.
C. RTTY.
D. AMTOR.

E2E09. Which of the following HF digital modes uses variable-length character coding?
A. RTTY.
B. PACTOR.
C. MT63.
D. PSK31.

E2E10. Which of these digital modes has the narrowest bandwidth?
A. MFSK16.
B. 170 Hz shift, 45-baud RTTY.
C. FT8.
D. PACTOR IV.

E2E11. What is the difference between direct FSK and audio FSK?
A. Direct FSK modulates the transmitter VFO.
B. Direct FSK occupies less bandwidth.
C. Direct FSK can transmit higher baud rates.
D. All these choices are correct.

E2E12. How do ALE stations establish contact?
A. ALE constantly scans a list of frequencies, activating the radio when the designated call sign is received.
B. ALE radios monitor an internet site for the frequency they are being paged on.
C. ALE radios send a constant tone code to establish a frequency for future use.
D. ALE radios activate when they hear their signal echoed by back scatter.

E2E13. Which of these digital modes has the highest data throughput under clear communication conditions?
A. MFSK16.
B. 170 Hz shift, 45 baud RTTY.
C. FT8.
D. PACTOR IV.

Answer Key

E2E01: B	E2E02: B	E2E03: B	E2E04: D
E2E05: A	E2E06: C	E2E07: C	E2E08: B
E2E09: D	E2E10: C	E2E11: A	E2E12: A
E2E13: D			